

PROGRAMMING IN HASKELL

0



Topic 8.2 Function Application

\$ as function application

1

\$ is the function application operator

$$(\$) :: (a \rightarrow b) \rightarrow a \rightarrow b$$
$$f \$ x = f x$$

\$ as function application

2

It's function application but:

- normal function application has high precedence, \$ has low precedence
- normal function application is left associate, e.g.,
 $f\ a\ b\ c === ((f\ a)\ b)\ c$
- \$ is right associative

\$ as function application

3

```
*Main> (^2) 4 + 3  
19
```

```
*Main> (^2) $ 4 + 3  
49
```


Improved syntax with \$

4

Most often it's a convenience that lets us write fewer parentheses.

Example:

```
sum (map sqrt [1..130])
```

Is better written as:

```
sum $ map sqrt [1..130]
```

when \$ is encountered, expression on right is used as parameter to function on left

More examples

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`sqrt (3+4+9)`  `sqrt $ 3+4+9`

`*Main> sum (filter (> 10) (map (*2) [2..10]))`

80

`*Main> sum $ filter (> 10) (map (*2) [2..10])`

80

`*Main> sum $ filter (> 10) $ map (*2) [2..10]`

80

Another example

6

```
*Main> (10*) $ 3
```

```
30
```

```
*Main> ($ 3) (10*)
```

```
30
```

```
map ($ 3) [(4+), (10*), (^2), sqrt]
```

```
[7.0,30.0,9.0,1.7320508075688772]
```

How does this work?

- expression on right is used as parameter to function on left

